

# Ariadne

federated knowledge for cybersecurity

# One question.
# Answers from many sources.

One analyst question, answered across security knowledge graphs that speak different languages. Every sentence carries its record.

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# What is a knowledge graph?

## Relational Database

Structured records

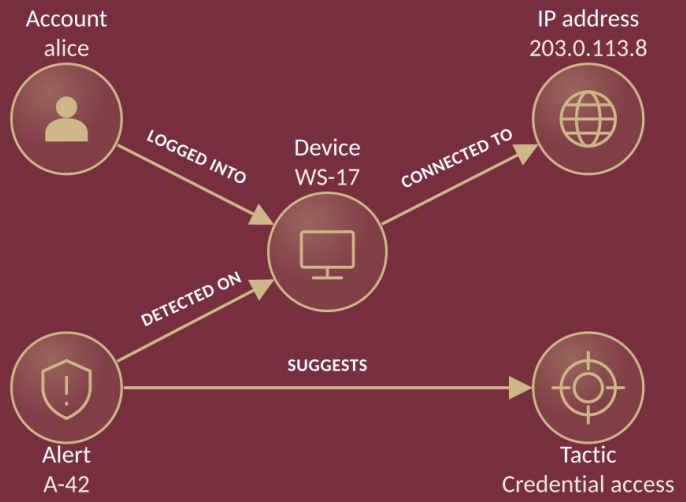
**Authentication events**

| Time | Account | Device | Result |
|---|---|---|---|
| 09:01 | alice | WS-17 | Failed |
| 09:02 | alice | WS-17 | Failed |
| 09:03 | bob | WS-22 | Failed |
| 09:04 | alice | WS-17 | Success |

How many failed logins per account?

## Knowledge Graph

Heterogeneous entities




What activity connects this account to a threat?

1. **Entities become nodes**
   an account, a device, an alert, a tactic
2. **Connections are named**
   logged into · connected to · detected on · suggests
3. **Questions become paths**
   follow the links instead of joining tables

**A table answers "how many". A graph answers "what connects".**

# Knowledge graphs speak different languages



## Cypher

Attacks

```
MATCH (a:Attack)
WHERE a.technique = 'T1003'
RETURN a.name
```

中文



## Gremlin

Mitigations

```
g.V().has('Mitigation',
  'technique', 'T1003')
 .values('name')
```

αβ



## GQL

Logs

```
MATCH (e:Event)
FILTER e.technique = 'T1003'
RETURN e.timestamp
```

あい

Connect attacks, mitigations, and logs across graphs.

1 **MITRE ATT&CK**

adversary techniques and mitigations

**Neo4j · speaks Cypher**

2 **MITRE CAPEC**

attack patterns

**Gremlin Server · speaks Gremlin**

3 **Cloud graph databases**

Spanner Graph, Amazon Neptune

**speak GQL and openCypher**

**Same question, three grammars.**

# How people query across graphs today

### 1 Write the query by hand

**Gives you** Accurate and auditable.

**Costs you** Slow, and it takes an expert per language, every time.

### 2 Ask a chatbot

**Gives you** Fast and easy to ask.

**Costs you** The query it ran is hidden, and a language model can invent facts.

### 3 Copy everything into one database

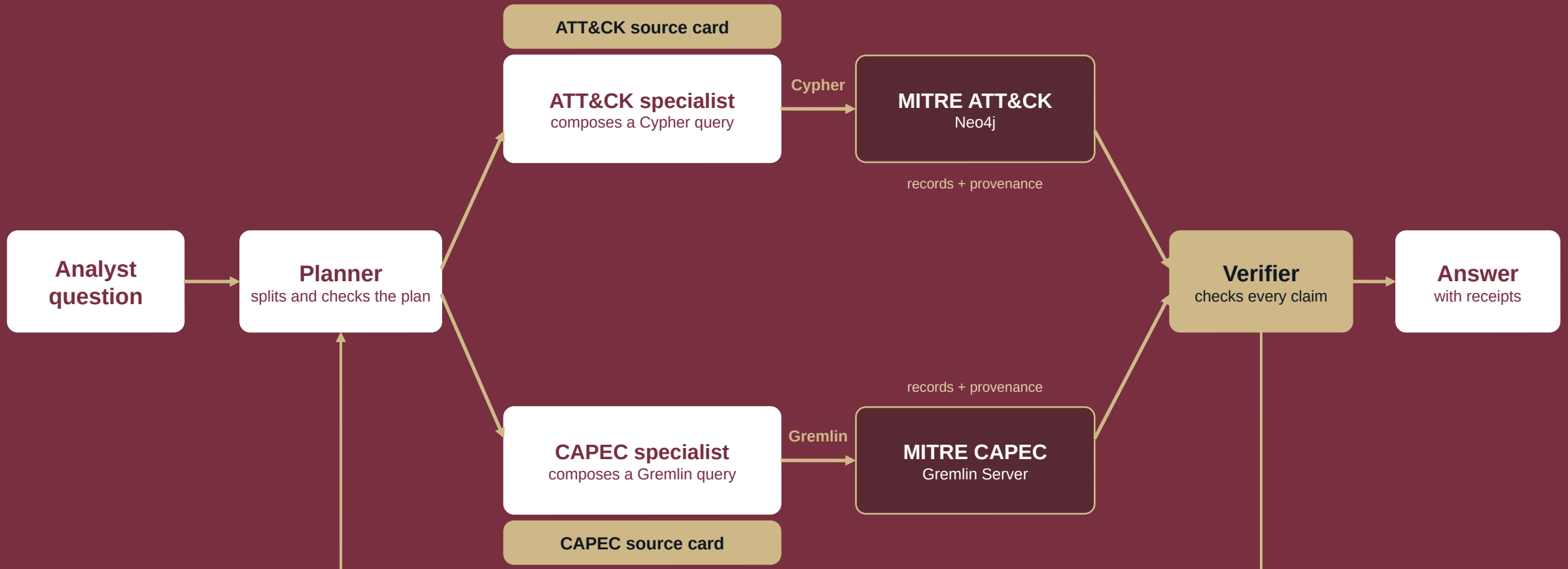
**Gives you** One language to learn.

**Costs you** Expensive, stale the day it lands, and it rewrites the sources.

***Hallucination:*** *when an AI states something as a fact that is not actually in the data.*

**Each one trades away accuracy, speed, or visibility.**

# How Ariadne answers one question




**Rule 1 · No model ever writes a query.**

**Rule 2 · No claim without a record behind it.**

# The source card: one sheet per database

`attack.json` the ATT&CK source card

- **Entities and properties**
  what the graph contains
- **Relationships**
  how entities connect
- **Join keys**
  how records link across databases
- **Query limits**
  read-only · depth · rows · timeout
- **Provenance**
  which snapshot, pinned by hash

**39 KB of card describes a 19.9 MB graph.**

## What the specialist does with it

1. **Resolve** the question's entities against the card
2. **Compose** the query from five schema-derived primitives
3. **Gate** it: read-only, allowed labels, bounded depth and rows
4. **Execute** once, under a timeout
5. **Normalize** every record with its provenance

# The verifier: every claim carries its record

- **Cited**
  every claim points at retrieved records
- **Resolvable**
  every identifier appears in those records
- **Grounded**
  what the claim says is in the record
- **Correlated**
  cross-database links are backed by the join index

**What fails is cut and flagged. Deterministic code, not a model.**

### When the data has nothing

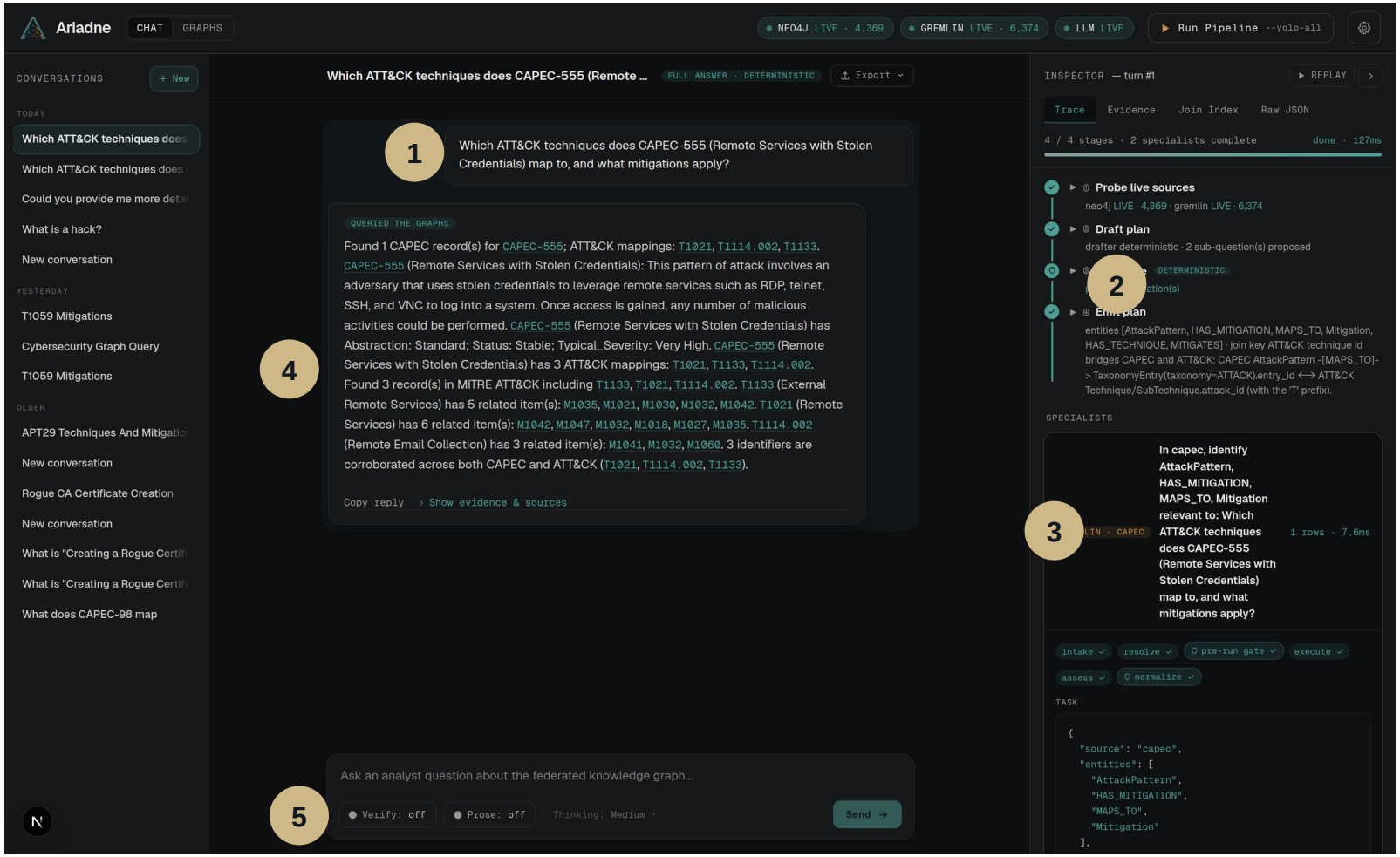
*"The capec source doesn't establish this. The executed query ran cleanly and returned no matching records."*

```
g.V().hasLabel("AttackPattern").has("id", within(ids))
 .out("MAPS_TO").hasLabel("TaxonomyEntry") …
```

**0 claims made · 0 invented mappings · the query is the proof**

An honest no, with the receipt.

# The workbench: ask, watch it run, open the receipts




1. **The question**
   plain English

2. **The plan and its gate**
   which sources, which sub-questions

3. **One lane per database**
   the validated query, in its own language

4. **The answer**
   every identifier is a citation

5. **Three controls, off by default**
   Verify · Prose · Thinking: model help, per question

# It scales across databases

## 3 · 2 · 1

**databases · query languages · one process**

Adding the third changed nothing about the questions.

## 0 vs 1,083

**records rewritten: federating vs merging**

Merging ATT&CK and CAPEC into one store rewrote 1,083 records. Federating rewrote none.

## 240 / 240

**adversarial queries blocked before execution**

After we found and closed 15 bypasses ourselves.

**Adding a database is a source card, not a project.**

Next: Amazon Neptune and Google Spanner Graph.

# It already runs on ReliaQuest's own schema

## What ReliaQuest sent

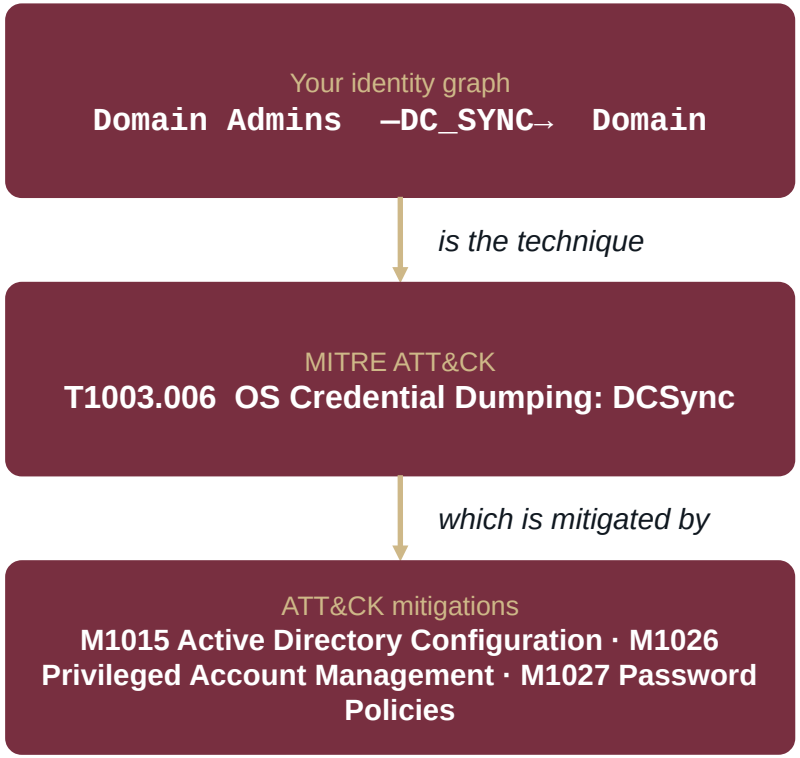
**72**
node types

**21**
relationship types

**5**
identity systems: AD · Entra · Okta · AWS · GCP

**0**
customer records needed

## The bridge to ATT&CK




13 of 21 relationship types map to a technique; the other 8 are declared not to.

## What happened

- The schema alone was enough. Records are synthetic; the schema is real.
- Ran live beside ATT&CK and CAPEC. Nothing else changed.
- A relationship in your graph resolves to a technique, then to its mitigations.
- Where no mapping exists, it says so.

***"Which technique is what Domain Admins can do with DC_SYNC, and what mitigates it?"***

# Where it stops today

| | | |
|---|---|---|
| 1 | **Ask by identifier**<br>"What mitigates T1003?" | **Works** |
| 2 | **Reword the question**<br>same entity, new phrasing | **About half** |
| 3 | **Paraphrase the name**<br>no identifier | **About a third** |
| 4 | **Describe it without naming it**<br>no identifier, no name | **Not yet** |

## Also open

- Answer quality through the workbench's own planner is not yet judged.
- Identity records were synthetic; the schema was real.
- The planner, not the engine, sets the ceiling. It is the next target.

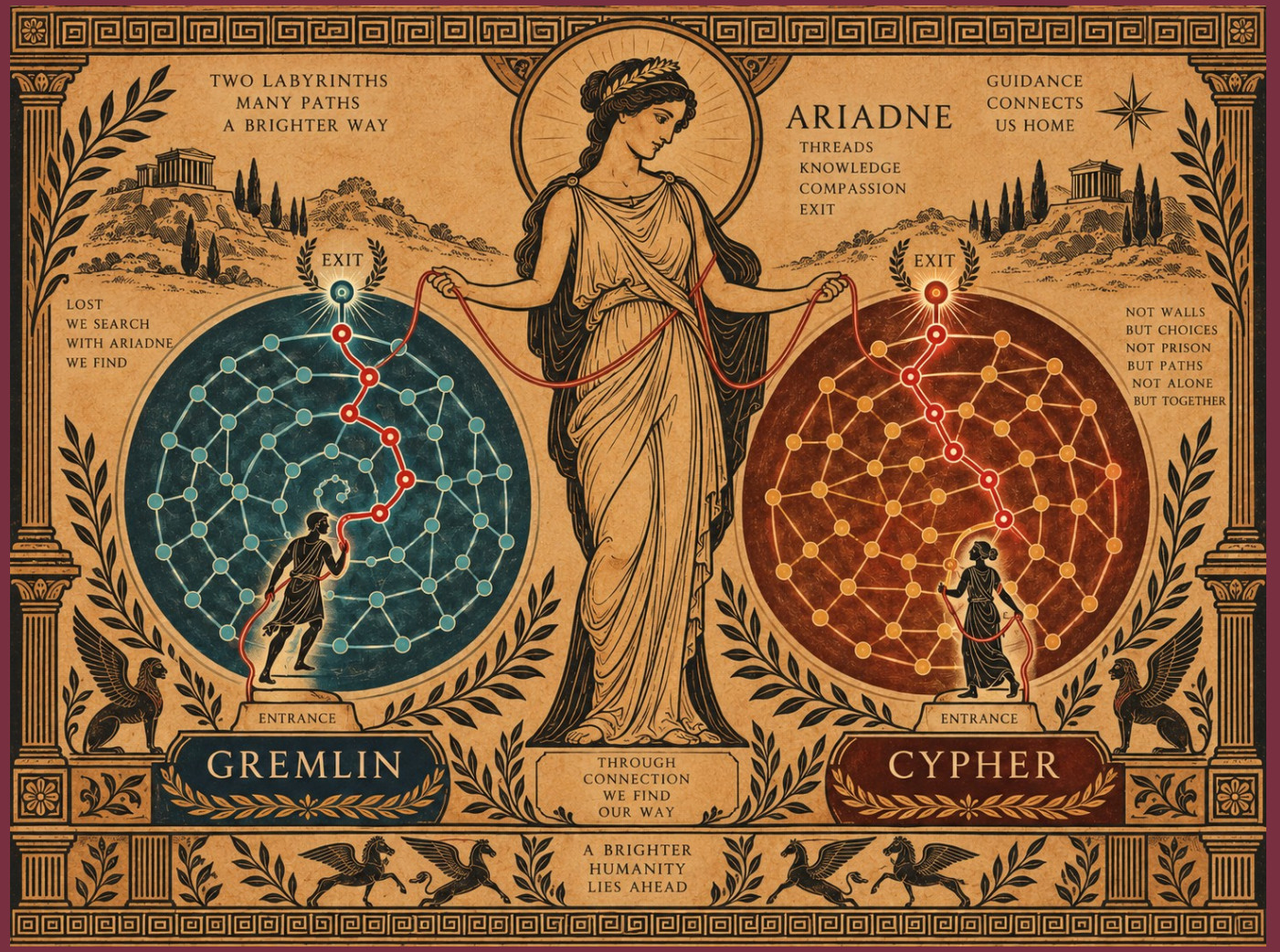
**We know the next gap precisely, and it is measurable.**

# Thank you

## Questions?

*One question. Answers from many sources.*

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github.com/brendondgr/Ariadne




*Ariadne's thread guides travelers through the Gremlin and Cypher labyrinths.*

# Works cited


## Data and tools

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## Prior and related work

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13. Zavala Romero, O. S., & DGR, B. (2026). Ariadne: federated agentic access to cybersecurity knowledge graphs. Research record, docs/research/, https://github.com/brendondgr/Ariadne


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